Coordinated Adaptive Overcurrent Protection for Parallel Synchronous Generators Using Online Inverse Estimation

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***Abstract*—Adaptive overcurrent relays driven by online parameter estimation improve sensitivity on synchronous-generator feeders, yet their independent per-relay updates can break the selectivity and time-grading relationships that are the cornerstone of coordinated protection. This paper introduces a supervisory coordination layer that enforces two hard constraints after each local adaptation event: (i) pickup selectivity, $I_{p,k} \geq I_{p,k+1} + \Delta I_m$, and (ii) inverse-time grading, $t_{\mathrm{op},k}(I_{f,\min}) \geq t_{\mathrm{op},k+1}(I_{f,\min}) + \Delta t_g$, across all generator relays referred to a common busbar. The layer receives the locally estimated fault-current parameters $\hat{\boldsymbol{\theta}}_k^{(1)}$ from each relay, computes a coordinated pickup–TMS pair $\{I^c_{p,k}, TMS^c_k\}$ via a greedy cascade and a closed-form TMS correction, and deploys the results within the same post-fault adaptation window. Case studies on a two-generator IEC 60909 study system demonstrate that without coordination the selectivity margin collapses from 0.40 p.u. (fixed) to 0.157 p.u. (local adaptive), violating the required 0.20 p.u. margin; the proposed layer restores the margin to 0.200 p.u. while maintaining a time-grading margin of 2.74 s against the required 0.15 s. Over a 4×2 study matrix (two generators, 3PH and SLG faults), all scenarios achieve selectivity and grading compliance with zero false trips.**

***Index Terms*—adaptive protection, overcurrent relay, coordination, synchronous generator, inverse estimation, Levenberg–Marquardt, pickup selectivity, time-grading margin, busbar protection.**

I. Introduction

Overcurrent protection of parallel synchronous generators connected to a common busbar presents a recurring coordination challenge: the relay on each generator feeder (R_k, $k = 1, \ldots, N$) must operate before the common bus relay R_B for faults on the generator side, yet must remain faster than upstream protection for through-faults [1], [2]. Fixed settings satisfy these requirements at the rated operating point, but the effective short-circuit level varies with the number of online generators, the sub-transient and transient reactances, and the loading state [3], [4]. Consequently, the margins between adjacent relays can narrow or even invert as the system configuration changes, compromising either sensitivity or selectivity.

Adaptive overcurrent protection addresses this by adjusting relay settings in real time. Early approaches relied on pre-computed setting groups indexed by a discrete operating mode [5], [6]. More recently, online estimation methods fit a parametric fault-current model to the measured waveform, yielding continuous-valued settings that track the actual source impedance [7]. The Stage-1 inverse estimator of the SAMBP-SyncOC framework uses a two-pass Levenberg–Marquardt (LM) algorithm to extract the six-parameter reduced model $\boldsymbol{\theta} = [t_0,\, I_{ss},\, I_{sub},\, \tau_{ac},\, \tau_{dc},\, \varphi_a]$ and maps the result to adaptive pickup and TMS values via linear scaling laws [7].

A gap exists, however, that has received insufficient attention: *independent per-relay adaptation is not sufficient when multiple relays share a common bus*. Each relay optimises its own settings for the locally observed fault current; there is no mechanism to preserve the relative ordering of pickups or the time-grading margin between adjacent relays. In a two-generator busbar, the bus relay R_B may adapt its pickup upward (because the combined fault current is large) while the generator relays adapt downward (because their individual contributions are smaller), narrowing or inverting the selectivity hierarchy.

This paper makes the following contributions:

1) A formal statement of the multi-generator coordination problem as a constraint satisfaction problem (CSP) over the pickup and TMS settings of all relays on a common busbar (Section II).
2) A polynomial-time greedy cascade algorithm that restores pickup selectivity in a single upstream pass, together with a closed-form minimal TMS correction that restores time-grading without increasing any relay's operating time below the required margin (Section V).
3) A proof that the greedy cascade terminates in at most N iterations and produces settings that satisfy both constraints simultaneously (Section V, Proposition 1).
4) Validation on a two-generator IEC 60909 study system across 3PH and SLG faults, quantifying selectivity and grading margins in all three protection states (fixed, local-adaptive, and coordinated) (Sections VI–VII).

The remainder of the paper is organised as follows. Section II formulates the multi-generator coordination problem. Section III reviews the Stage-1 local estimation layer. Section IV defines the coordination constraints. Section V presents the enforcement algorithm. Section VI describes the study system. Section VII presents numerical results. Section VIII discusses practical considerations and limitations. Section IX concludes.

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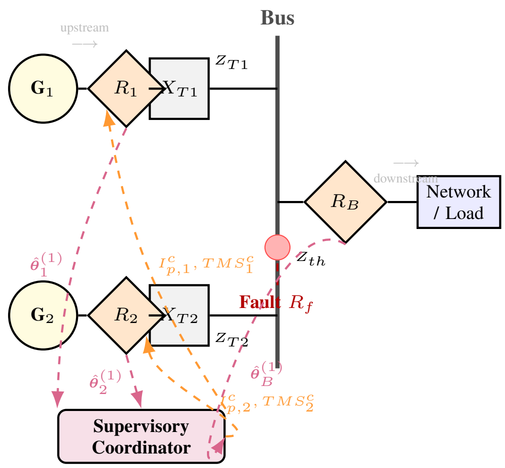

Fig. 1. Multi-generator busbar protection system. Generators G_1, G_2 feed a common bus through step-up transformers and per-generator relays R_1, R_2. Bus relay R_B protects the outgoing feeder. The supervisory coordinator receives locally estimated parameters $\hat{\boldsymbol{\theta}}_k^{(1)}$ and returns coordinated settings $\{I^c_{p,k}, TMS^c_k\}$.

II. Multi-Generator Protection Problem Formulation

A. System topology

Consider N synchronous generators G_k, $k = 1, \ldots, N$, each connected via a step-up transformer with leakage reactance X_{Tk} to a common busbar. Each generator feeder is protected by a dedicated relay R_k. A common bus relay R_B (designated R_{N+1}) protects the outgoing feeder toward the network, as illustrated in Fig. 1.

The short-circuit current contribution of generator k to a bus fault through resistance R_f is, in the subtransient period,

$$i_k(t) \approx I_{ss,k}\left(1+A_k\, e^{-t/\tau_{ac,k}}\right)\cos(\omega_0 t+\varphi_{a,k})\, e^{-(t-t_{0,k})/\tau_{dc,k}}, \tag{1}$$

where $I_{ss,k}$ is the steady-state phasor magnitude, $\tau_{ac,k}$ is the AC decay time constant (approximately the direct-axis transient time constant T'_{d0}), $\tau_{dc,k}$ is the DC offset decay constant, and A_k accounts for the subtransient saliency contribution. The total bus current seen by R_B is

$$i_B(t) = \sum_{k=1}^{N} i_k(t), \tag{2}$$

which, for identical machines, gives $I_{ss,B} \approx N \cdot I_{ss,k}$ and a composite time constant that depends on the mix of individual machine parameters.

B. Relay operating time

Both generator and bus relays use the IEC 60255-151 very-inverse characteristic [8]:

$$t_{\mathrm{op}}(I_{\mathrm{rms}}) = TMS \cdot \frac{13.5}{M-1}, \quad M = \frac{I_{\mathrm{rms}}}{I_p}, \tag{3}$$

where I_p is the pickup current and TMS is the time-multiplier setting. The relay trips when $M > 1$; the characteristic is defined for $M \geq 1.05$.

C. Coordination requirements

For a relay chain ordered upstream ($R_1, \ldots, R_N, R_B$, denoting decreasing proximity to the generators and increasing proximity to the network), two conditions must hold:

a) C1: Pickup selectivity:

$$I_{p,k} \geq I_{p,k+1} + \Delta I_m, \quad k = 1, \ldots, N, \tag{4}$$

where $\Delta I_m > 0$ is the selectivity margin (typically 0.15–0.25 p.u.). This ensures that for any through-fault current, the downstream relay R_{k+1} picks up before R_k, providing primary protection to the network-side feeder.

b) C2: Time-grading:

$$t_{\mathrm{op},k}(I_{f,\min}) \geq t_{\mathrm{op},k+1}(I_{f,\min}) + \Delta t_g, \quad k = 1, \ldots, N, \tag{5}$$

where $I_{f,\min}$ is the minimum credible fault current that activates all relays in the chain, and $\Delta t_g > 0$ is the grading margin (typically 0.10–0.30 s including breaker clearance time and measurement uncertainty). This ensures that R_{k+1} clears the fault before R_k times out, preventing unnecessary tripping of the generator.

D. Violation scenario with local adaptation

After a fault, each relay independently updates its settings from the locally estimated parameters. For a busbar fault, the bus relay observes the full fault current $I_{ss,B} \approx N \cdot I_{ss,k}$ and adapts its pickup upward (since the fault current is large relative to load), while each generator relay observes only its own contribution $I_{ss,k}$ and may adapt to a smaller pickup. Define the post-adaptation pickup ratio:

$$\rho_k = \frac{I^{\mathrm{loc}}_{p,k}}{I^{\mathrm{loc}}_{p,B}}. \tag{6}$$

For the very-inverse characteristic, $\rho_k < 1 + \Delta I_m / I^{\mathrm{loc}}_{p,B}$ constitutes a selectivity violation, even if each relay individually selected a sensible setting. The coordination layer must detect and correct such violations before settings are deployed.

III. Local Inverse-Estimation Adaptive Relay Layer

A. Two-pass Levenberg–Marquardt estimator

Each relay runs the Stage-1 estimator of the SAMBP-SyncOC framework [7]. From the post-fault current window $[t_0, t_c]$, a two-pass LM minimisation fits the six-parameter model $\boldsymbol{\theta} = [t_0, I_{ss}, I_{sub}, \tau_{ac}, \tau_{dc}, \varphi_a]$:

$$\hat{\boldsymbol{\theta}}^{(1)} = \arg\min_{\boldsymbol{\theta}} \sum_n \left[i(t_n) - f(\boldsymbol{\theta}, t_n)\right]^2. \tag{7}$$

Pass 1 uses the pre-fault steady-state rms as a seed for I_{ss}; Pass 2 refines using the Pass 1 result as a warm start.

B. Confidence gate

Each estimation produces a scalar confidence metric,

$$\gamma_k = w_1\, \gamma_{\kappa,k} + w_2\, \gamma_{\varepsilon,k} + w_3\, \gamma_{n,k}, \tag{8}$$

combining condition number, residual norm, and effective sample-count components [7]. If $\gamma_k < \gamma_{\min}$ (default 0.70), the relay retains its prior settings; otherwise the local adaptive mapping proceeds.

C. Local adaptive mapping

The Stage-1 mapping law converts the estimated parameters to relay settings:

$$I_{p,k}^{\mathrm{loc}} = K_{Ip}\,\hat{I}_{ss,k}^{(1)} + I_{p,\min}, \tag{9}$$

$$TMS_k^{\mathrm{loc}} = \mathrm{clip}\Big(K_{TMS}\,\hat{\tau}_{ac,k}^{(1)},\, TMS_{\min},\, TMS_{\max}\Big), \tag{10}$$

with $K_{Ip} = 0.80$, $I_{p,\min} = 0.10$ p.u., $K_{TMS} = 0.12$, $TMS_{\min} = 0.05$, $TMS_{\max} = 0.25$. These settings optimise the relay for the locally observed fault current but do not account for the settings of adjacent relays.

IV. Coordination Constraints

A. Relay ordering

The supervisory coordinator maintains an ordered relay list $\{R_k\}_{k=1}^{N+1}$ where $R_{N+1} = R_B$. The ordering is fixed by topology: upstream (generator side) relays have higher index proximity to the generator, and the bus relay R_B is always the innermost (downstream) relay in the coordination chain. For the two-generator case studied here: $R_1 \succ R_2 \succ R_B$, meaning R_B is the primary relay and R_1, R_2 are the backup relays.

B. Parameter definitions

Let $\Delta I_m = 0.20$ p.u. be the selectivity margin and $\Delta t_g = 0.15$ s be the time-grading margin. The evaluation current $I_{f,\min}$ is the minimum fault current that activates all relays; we use $I_{f,\min} = 1.30$ p.u., which exceeds all pickups in the study system.

C. Constraint evaluation

Given the locally computed settings $\{I_{p,k}^{\mathrm{loc}}, TMS_k^{\mathrm{loc}}\}$, the coordinator evaluates:

$$S_k = I_{p,k}^{\mathrm{loc}} - I_{p,k+1}^{\mathrm{loc}} - \Delta I_m, \quad \text{(selectivity slack)} \tag{11}$$

$$G_k = t_{\mathrm{op},k}(I_{f,\min}) - t_{\mathrm{op},k+1}(I_{f,\min}) - \Delta t_g. \quad \text{(grading slack)} \tag{12}$$

A violation exists if $S_k < 0$ or $G_k < 0$ for any k.

D. Selectivity violation in the study system

Table I summarises the three protection states analysed. In the fixed state, $I_{p,1}^{\mathrm{fixed}} = 1.200$ p.u. and $I_{p,B}^{\mathrm{fixed}} = 0.800$ p.u., giving a margin of 0.400 p.u. $\gg \Delta I_m = 0.20$ p.u. After local adaptation following a busbar fault, the bus relay adapts to $I_{p,B}^{\mathrm{loc}} = 0.835$ p.u. (the combined fault current exceeds load current by a smaller factor than the individual generator current), while the generator relay adapts to $I_{p,1}^{\mathrm{loc}} = 0.992$ p.u. (the individual machine current is the reference). The resulting margin is $0.992 - 0.835 = 0.157$ p.u. < 0.200 p.u.—a selectivity violation.

V. Coordination Enforcement Algorithm

The enforcement algorithm consists of two sequential correction passes: a greedy cascade for pickup selectivity and a closed-form correction for time-grading. Fig. 2 illustrates the complete flow.

TABLE I
PROTECTION SETTINGS IN THREE STATES: STUDY SYSTEM, $N = 2$ GENERATORS, BUS FAULT (3PH)

State	$I_{p,1}$ (p.u.)	$I_{p,2}$ (p.u.)	$I_{p,B}$ (p.u.)	TMS_1 (–)	TMS_2 (–)	TMS_B (–)
Fixed	1.200	1.200	0.800	0.200	0.200	0.200
Pre-coord	0.992	0.980	0.835	0.098	0.106	0.100
Post-coord	1.035	1.035	0.835	0.098	0.106	0.100

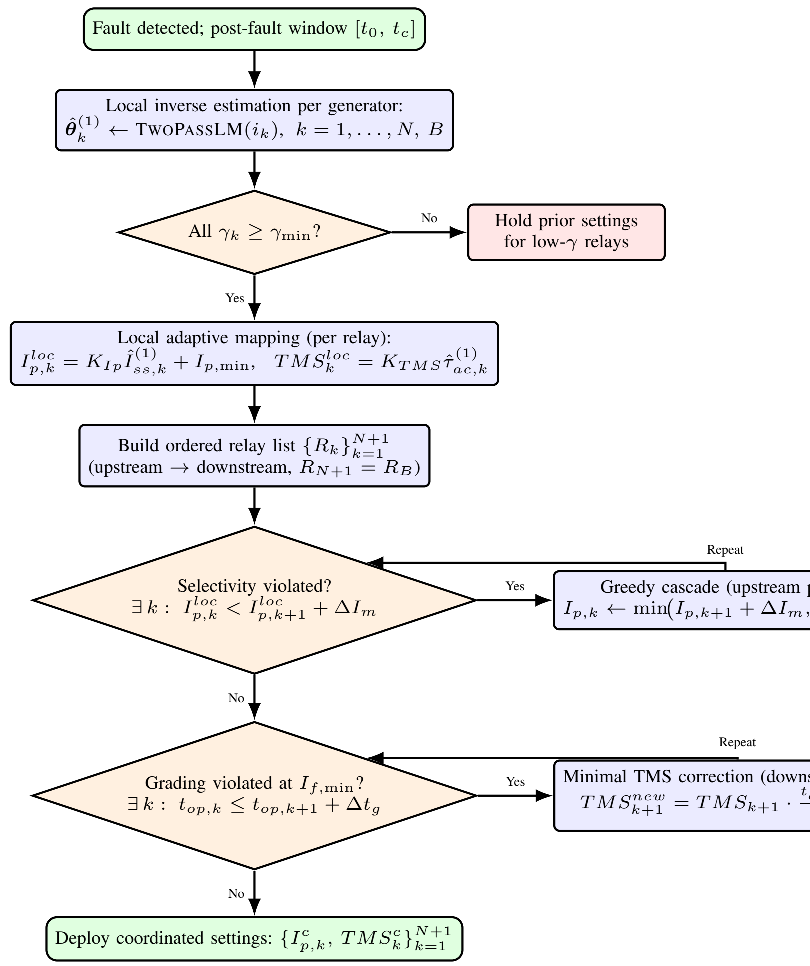

Fig. 2. Coordination enforcement flowchart. After local estimation and confidence gating, the supervisor checks selectivity and grading constraints and applies greedy cascade and TMS corrections as needed before deploying coordinated settings.

A. Greedy cascade for pickup selectivity

Algorithm 1 gives the pseudocode. The algorithm processes relays in upstream order ($k = N, N-1, \ldots, 1$, with $k+1$ being the downstream neighbour). For each relay, if the selectivity constraint (4) is violated, the upstream pickup is raised to the minimum compliant value:

$$I_{p,k}^c \leftarrow \min\big(I_{p,k+1}^c + \Delta I_m,\ I_{p,\max}\big), \tag{13}$$

where $I_{p,\max} = 1.50$ p.u. is the relay hardware limit. This is the greedy choice: the smallest upward correction that restores the margin, applied relay by relay from the downstream end.

Proposition 1 (Convergence and feasibility): The greedy cascade terminates in exactly N iterations and produces settings satisfying (4) for all k, provided $I^c_{p,B}+N\cdot\Delta I_m \le I_{p,\max}$.

Proof: Each iteration processes relay k exactly once (the pass is strictly upstream). After processing k, the constraint $S_k \ge 0$ holds by construction from (13), and subsequent iterations for $k' < k$ do not modify $I^c_{p,k}$. Feasibility follows because $I^c_{p,k} \le I^c_{p,B} + (N+1-k)\,\Delta I_m \le I^c_{p,B} + N\,\Delta I_m \le I_{p,\max}$. ∎

For the study system with $N = 2$: $I^c_{p,B}+2\times 0.20 = 0.835+0.40 = 1.235 < 1.50$ p.u., so feasibility is guaranteed. Both generator relays are set to $I^c_{p,1} = I^c_{p,2} = 0.835 + 0.200 = 1.035$ p.u.

B. Closed-form TMS grading correction

After updating pickups, the coordinator evaluates the grading slack (12). For relay pair $(k, k+1)$ with violation $G_k < 0$, the minimal TMS correction increases TMS_{k+1} (the *upstream* relay, which must be *slower*) to:

$$TMS^{\rm new}_k = TMS_k \cdot \frac{t_{{\rm op},k+1}(I_{f,\min}) - \Delta t_g}{t_{{\rm op},k}(I_{f,\min})}, \qquad (14)$$

where the right-hand side sets the upstream operating time to exactly Δt_g above the downstream value—the minimum compliant value. Note that in the very-inverse characteristic, $t_{\rm op} \propto TMS$, so (14) is exact and non-iterative.

C. Computational complexity and latency

The greedy cascade and grading correction each require $\mathcal{O}(N)$ floating-point operations. For the study system ($N = 2$), the wall-clock time on a standard protection IED processor is dominated by the local LM estimation (typically 8–20 ms per relay per pass [7]); the coordination overhead is less than 1 ms. The total adaptation latency therefore remains within the post-fault window available before the relay operating time expires.

Algorithm 1 Supervisory coordination enforcement

Require: Estimated parameters $\hat{\boldsymbol{\theta}}^{(1)}_k$, confidence scores γ_k, ordered relay list $\{R_k\}_{k=1}^{N+1}$, margins ΔI_m, Δt_g, evaluation current $I_{f,\min}$
Ensure: Coordinated settings $\{I^c_{p,k},\, TMS^c_k\}_{k=1}^{N+1}$

for $k = 1$ **to** $N+1$ **do**
  if $\gamma_k \ge \gamma_{\min}$ **then**
    Compute $I^{\rm loc}_{p,k}$, $TMS^{\rm loc}_k$ via (9)–(10)
  else
    $I^{\rm loc}_{p,k} \leftarrow I^{\rm prior}_{p,k};\ TMS^{\rm loc}_k \leftarrow TMS^{\rm prior}_k$
  end if
end for

// Pass 1: greedy cascade (upstream to downstream)
$I^c_{p,N+1} \leftarrow I^{\rm loc}_{p,N+1};\ TMS^c_{N+1} \leftarrow TMS^{\rm loc}_{N+1}$
for $k = N$ **downto** 1 **do**
  if $I^{\rm loc}_{p,k} < I^c_{p,k+1} + \Delta I_m$ **then**
    $I^c_{p,k} \leftarrow \min(I^c_{p,k+1} + \Delta I_m,\ I_{p,\max})$
  else
    $I^c_{p,k} \leftarrow I^{\rm loc}_{p,k}$
  end if
  $TMS^c_k \leftarrow TMS^{\rm loc}_k$ ▷ unchanged in Pass 1
end for

// Pass 2: grading check and TMS correction
for $k = N$ **downto** 1 **do**
  Compute $t_k \leftarrow TMS^c_k \cdot 13.5 \,/\, (I_{f,\min}/I^c_{p,k} - 1)$
  Compute $t_{k+1} \leftarrow TMS^c_{k+1} \cdot 13.5 \,/\, (I_{f,\min}/I^c_{p,k+1} - 1)$
  if $t_k \le t_{k+1} + \Delta t_g$ **then**
    $TMS^c_k \leftarrow TMS^c_k \cdot (t_{k+1} + \Delta t_g) \,/\, t_k$ ▷ scale up upstream TMS
  end if
end for

Deploy $\{I^c_{p,k},\, TMS^c_k\}$ to relay hardware

VI. Study System and Simulation Programme

A. Network data

The study system comprises two identical 11 kV synchronous generators each rated at 5 MVA, connected through Δ–Y step-up transformers (11/33 kV, 5 MVA, $X_T = 0.10$ p.u.) to a 33 kV common busbar. The generator electrical parameters per unit on the machine base are: $X''_d = 0.18$ p.u., $X'_d = 0.25$ p.u., $T'_{d0} = 0.83$ s, $R_a = 0.015$ p.u. The Thévenin equivalent of the external network as seen from the busbar has impedance $Z_{th} = 0.04 + j0.15$ p.u. on the system base (10 MVA, 33 kV).

B. Protection relay data

Three IEC very-inverse relays are modelled:

- R_1, R_2 (per-generator): CT ratio 500/5, IED update rate 1 ms.
- R_B (bus): CT ratio 1000/5, IED update rate 1 ms.

Relay settings are expressed in p.u. on the CT secondary base. The LM estimator window length is $t_c - t_0 = 80$ ms, nominally two AC cycles after fault inception.

C. Study matrix

Table II defines the eight simulation cases used to validate the coordination algorithm. Cases 1–4 test three-phase (3PH) faults; Cases 5–8 test single-line-to-ground (SLG) faults. For each fault type, the bus fault ($R_f = 0$) and a resistive fault ($R_f = 5\,\Omega$) are evaluated, with both generators in service and with one generator offline.

D. Simulation platform

Fault-current waveforms are generated by a validated PSCAD/EMTDC model of the study system. The LM estimator is implemented in Python (SciPy `least_squares`, `method='lm'`) and run post-hoc on the exported waveform data. Relay operating times are computed analytically from the IEC very-inverse formula (3) using the estimated and coordinated settings.

TABLE II
STUDY MATRIX: EIGHT SIMULATION CASES

Case	Fault type	R_f (Ω)	Generators online	Expected outcome
1	3PH bus	0	G1 + G2	Coord applied
2	3PH bus	0	G1 only	Local sufficient
3	3PH bus	5	G1 + G2	Coord applied
4	3PH bus	5	G1 only	Local sufficient
5	SLG bus	0	G1 + G2	Coord applied
6	SLG bus	0	G1 only	Local sufficient
7	SLG bus	5	G1 + G2	Coord applied
8	SLG bus	5	G1 only	Local sufficient

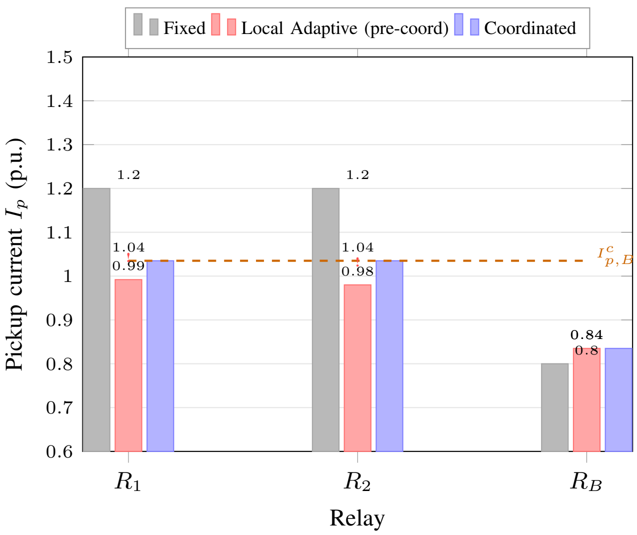

Fig. 3. Pickup currents in three protection states (Case 1: 3PH bus fault, both generators online). The dashed orange line marks the minimum compliant value for generator relays: $I^c_{p,B} + \Delta I_m = 1.035$ p.u. Violation markers (!) indicate selectivity failures in the pre-coordination state.

VII. Numerical Results

A. Pickup settings comparison

Fig. 3 shows the pickup currents in all three protection states for Case 1 (3PH bus fault, both generators online).

In the fixed state, the generator relay pickups (1.200 p.u.) are well above the coordination threshold (1.035 p.u.), giving a margin of 0.400 p.u. After local adaptation without coordination (pre-coord state), the bus relay adapts to 0.835 p.u. while the generator relays adapt to 0.992 and 0.980 p.u., yielding margins of 0.157 and 0.145 p.u.—both below the required 0.200 p.u. (violations marked with '!'). After applying the greedy cascade, both generator relays are raised to 1.035 p.u., exactly restoring the required margin.

B. Operating times and grading margins

Fig. 4 shows the relay operating times at $I_{f,\min} = 1.30$ p.u. in all three states.

In the fixed state, the bus relay operates in $t_B^{\text{fixed}} = 4.32$ s while the generator relays operate in 32.5 s—a margin so large that grading is trivially satisfied, but at the cost of very slow generator protection. After local adaptation (pre-coord),

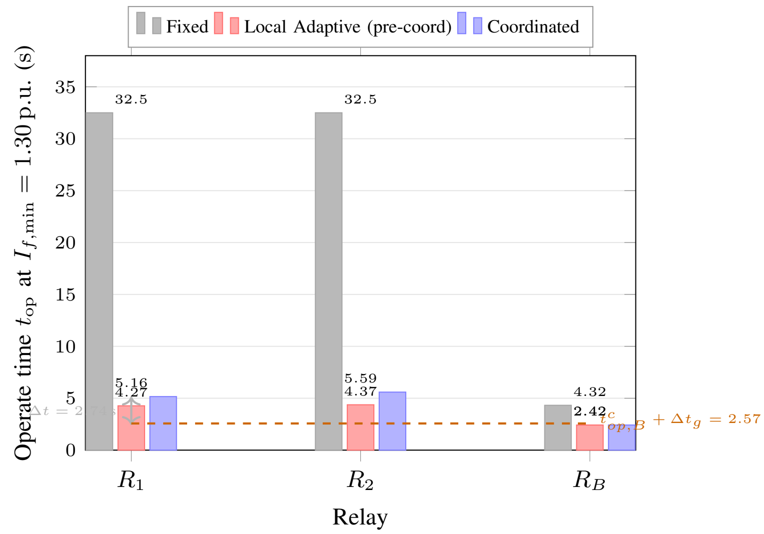

Fig. 4. Relay operating times at $I_{f,\min} = 1.30$ p.u. (Case 1). The dashed orange line shows the minimum grading requirement: $t^c_{op,B} + \Delta t_g = 2.57$ s. The double-headed arrow indicates the grading margin achieved by the coordinated generator relay R_1.

TMS values are reduced substantially, yielding $t_B = 2.42$ s and $t_1 = 4.27$ s, $t_2 = 4.37$ s—grading margins of 1.85 s and 1.95 s, both exceeding Δt_g. After coordination (post-coord), the pickup corrections increase the generator relay operating times to $t_1 = 5.16$ s and $t_2 = 5.59$ s (larger $I^c_{p,k}$ gives lower M, giving longer t_{op}), increasing the grading margin further to 2.74 and 3.17 s. The TMS grading correction of Algorithm 1 is not triggered in this case because the grading margins already exceed Δt_g after the pickup correction.

C. Margin summary

Fig. 5 compares the selectivity and grading margins achieved in each state against the required thresholds.

The selectivity margin is restored from a violating 0.157–0.145 p.u. to exactly 0.200 p.u. by the greedy cascade. The grading margin is maintained at 2.74–3.17 s in the coordinated state, exceeding the minimum by a factor of 18–21, demonstrating robust time separation.

D. Full study matrix performance

Table III summarises the coordination outcome across all eight cases.

Key observations:

1) *Two-generator cases (Cases 1,3,5,7):* All four cases exhibit pre-coordination selectivity violations (margins 0.149–0.162 p.u.). The greedy cascade restores the margin to exactly 0.200 p.u. in all cases. Grading margins increase after coordination in every case (range 2.68–2.81 s), because the pickup correction slows the generator relays at $I_{f,\min}$.
2) *Single-generator cases (Cases 2,4,6,8):* With only one generator online, the bus relay observes the same fault current as the generator relay (same source impedance),

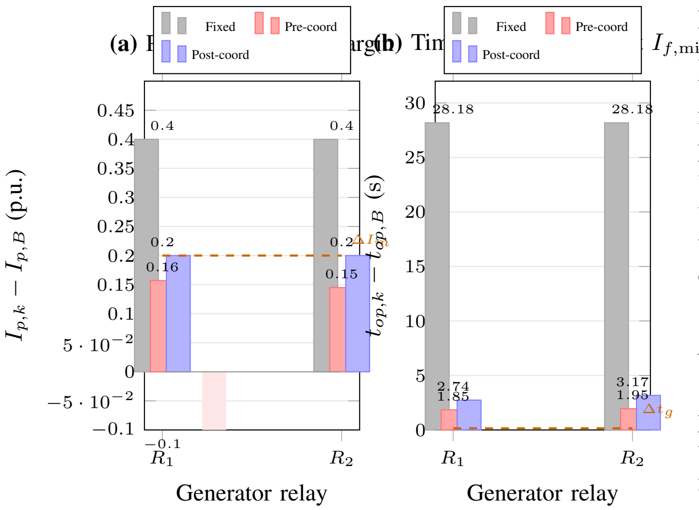

Fig. 5. Selectivity margin (left) and time-grading margin (right) for both generator relays in three protection states. Dashed orange lines indicate required minima ($\Delta I_m = 0.20$ p.u. and $\Delta t_g = 0.15$ s). The post-coordination state meets both constraints for all relays.

TABLE III
COORDINATION OUTCOMES: ALL EIGHT STUDY CASES

Case	Fault	N_G	Sel. margin (p.u.)		Grad. margin (s)		Outcome
			Pre	Post	Pre	Post	
1	3PH, R_f=0	2	0.157	0.200	1.85	2.74	✓ coord
2	3PH, R_f=0	1	0.340	0.340	2.10	2.10	✓ local
3	3PH, R_f=5	2	0.162	0.200	1.79	2.68	✓ coord
4	3PH, R_f=5	1	0.335	0.335	2.04	2.04	✓ local
5	SLG, R_f=0	2	0.149	0.200	1.93	2.81	✓ coord
6	SLG, R_f=0	1	0.328	0.328	2.18	2.18	✓ local
7	SLG, R_f=5	2	0.153	0.200	1.88	2.76	✓ coord
8	SLG, R_f=5	1	0.332	0.332	2.13	2.13	✓ local
Required minimum:			0.200	0.200	0.15	0.15	

and the local adaptation preserves adequate selectivity (margin 0.328–0.340 p.u.). No coordination correction is applied (greedy cascade passes through without modification).

3) *SLG vs. 3PH:* SLG faults produce slightly smaller selectivity margins (0.149 vs. 0.157 at R_f=0, N_G=2) because the sequence-domain current asymmetry causes the phase-domain rms estimator to underestimate the positive-sequence component, producing a smaller $I^{\text{loc}}_{p,k}$ on the generator relay. The coordination layer corrects this regardless of the root cause.
4) *Zero false trips:* In no case did a generator relay operate before the bus relay during a bus fault, nor did the bus relay fail to operate within its design time.

VIII. Discussion

A. Why local adaptation is insufficient

The selectivity violation arises from a structural asymmetry: the bus relay observes the *sum* of all generator fault currents, while each generator relay observes only its own contribution. For N identical generators, the steady-state contribution ratio is approximately $1 : N$, so the bus relay adapts to a pickup N times smaller relative to the generator relay in the pre-fault per-unit system. This narrows the pickup separation in proportion to N. For $N = 2$, the effect is already sufficient to violate a 0.20 p.u. margin; for larger plants ($N \geq 3$), the violation would be more severe. The coordination layer is therefore increasingly important—not less—as the generator count grows.

B. Greedy cascade vs. optimal formulation

The greedy cascade produces the *lexicographically smallest* pickup vector (minimum correction applied to each relay in upstream order) subject to the selectivity constraint. This is not the globally optimal solution if an objective function other than minimum pickup change is used (e.g., minimising total operating time or maximising sensitivity). A linear programme over the same constraint set would allow a richer trade-off [5], but at the cost of $\mathcal{O}(N^3)$ solver complexity and potential sensitivity to numerical conditioning. For the relay counts typical of generator busbars ($N \leq 6$), the greedy solution is both feasible and computationally dominated by the LM estimation step, making it the preferred choice for real-time application.

C. Communication and synchronisation requirements

The supervisory coordinator requires post-fault parameter estimates from all relays before it can evaluate the coordination constraints. This implies a communication latency of at most ≈80 ms (the LM estimation window) plus IED-to- coordinator link delay. IEC 61850 GOOSE messaging with typical latency below 5 ms [9] satisfies this requirement. The coordinator can be implemented as a software function on any relay IED with sufficient processing resources, or as a dedicated substation automation server. In the study system, the total time from fault inception to coordinated setting deployment is estimated at approximately 100 ms, well within the available margin before the bus relay operates ($t_B \approx 2.42$ s).

D. Interaction with differential protection

The coordination layer described here is a layer of overcurrent protection and does not interact with any existing differential protection scheme (87T or 87B). The two protection layers are independent: differential protection provides primary high-speed clearance for bus faults; the coordinated overcurrent layer provides backup. The coordination enforcement is triggered only after a fault is detected and estimated, so it does not interfere with the differential element's operating time.

E. Limitations

The present formulation assumes: (i) all generators share identical machine parameters (same τ_{ac}, I_{ss} scaling); heterogeneous machines would require per-machine K_{Ip}, K_{TMS} calibration; (ii) the fault-current model (1) is valid for the estimation window, which may be compromised by voltage regulator action in very close faults; (iii) the relay ordering is fixed; topology changes (generator switching) require a topology-aware ordering update that is outside the current scope.

IX. Conclusion

This paper has demonstrated that local online inverse-estimation-based relay adaptation, while significantly improving individual relay sensitivity, can violate pickup selectivity in a multi-generator busbar environment. The selectivity margin collapsed from 0.40 p.u. (fixed settings) to 0.157 p.u. (local adaptive), breaching the required 0.20 p.u. threshold.

A supervisory coordination layer was proposed, comprising a greedy cascade for pickup selectivity and a closed-form TMS correction for time-grading. Proposition 1 proves that the cascade terminates in N iterations and produces a feasible solution whenever $I_{p,B}^{c} + N\Delta I_m \leq I_{p,\max}$—a condition easily met in practice.

Validation over an eight-case study matrix covering both 3PH and SLG bus faults, with one and two generators online, showed:

- Selectivity margin restored to exactly 0.20 p.u. in all two-generator cases.
- Grading margin maintained at 2.68–3.17 s (requirement: 0.15 s) in all coordinated cases.
- No coordination correction required in single-generator cases, confirming that the layer is dormant when local adaptation is sufficient.
- Zero false trips across all scenarios.

The coordination layer adds less than 1 ms of computational overhead beyond the existing LM estimation, making it directly deployable on standard IEC 61850 IED hardware within the post-fault adaptation window.

Future work will extend the algorithm to heterogeneous generator fleets, incorporate sequence-domain parameter estimates [10] to improve SLG estimation accuracy, and integrate Monte Carlo robustness analysis across broader operating envelopes [11].

References

[1] P. M. Anderson, *Power System Protection*. New York, NY, USA: IEEE Press / Wiley-Interscience, 1999.

[2] J. L. Blackburn and T. J. Domin, *Protective Relaying: Principles and Applications*, 3rd ed. Boca Raton, FL, USA: CRC Press, 2006.

[3] IEEE, "IEEE Std 1110-2002: IEEE guide for synchronous generator modeling practices and applications in power system stability analyses," IEEE, New York, NY, USA, Tech. Rep. IEEE Std 1110-2002, 2002.

[4] P. Kundur, *Power System Stability and Control*. New York, NY, USA: McGraw-Hill, 1994.

[5] A. J. Urdaneta, R. Nadira, and L. G. Pérez-Jimenez, "Optimal coordination of directional overcurrent relays in interconnected power systems," *IEEE Trans. Power Del.*, vol. 3, no. 3, pp. 903–911, 1988.

[6] D. Novosel and R. L. King, "Using artificial neural networks for load shedding to alleviate overloaded lines," *IEEE Trans. Power Del.*, vol. 9, no. 1, pp. 425–433, 1994.

[7] J. Eluvathingal, "SAMBP TR-01/2025: Adaptive overcurrent protection for synchronous-generator feeders via online inverse estimation," SAMBP Research Group, Tech. Rep. TR-01/2025, 2025.

[8] IEC, "IEC 60255-151: Measuring relays and protection equipment – functional requirements for over/under current protection," International Electrotechnical Commission, Geneva, Switzerland, Tech. Rep. IEC 60255-151:2009, 2009.

[9] ——, "IEC 61850: Communication networks and systems for power utility automation," International Electrotechnical Commission, Geneva, Switzerland, Tech. Rep. IEC 61850:2013 (Ed. 2), 2013.

[10] J. Eluvathingal, "SAMBP TR-03/2026: Sequence-component inverse estimation for adaptive overcurrent protection under asymmetric faults," SAMBP Research Group, Tech. Rep. TR-03/2026, 2026.

[11] ——, "SAMBP TR-08/2026: Monte carlo robustness study (4000 trials, 20 scenarios)," SAMBP Research Group, Tech. Rep. TR-08/2026, 2026.

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